PROJECT REPORT

AGRICULTURE CROP RECOMMENDATION SYSTEM

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PROJECT CONTENT

A web-based Agriculture Crop Recommendation System that helps farmers select suitable crops based on location, season, water availability, and other agricultural information.

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Workbook Template

Project Name:

Agriculture Crop Recommendation System (Crop Care)

Problem Statement

Farmers often face difficulties in selecting the most suitable crop for cultivation because crop growth depends on factors such as soil conditions, temperature, rainfall, humidity and other environmental conditions. Choosing an unsuitable crop may result in low production and financial loss. An Agriculture Crop Recommendation System can help farmers select suitable crops based on available agricultural and environmental information.

Why does this problem exist?

1. Farmers may not have enough information about suitable crops.
2. Soil conditions differ from one location to another.
3. Weather conditions can change over time.
4. Different crops require different soil and environmental conditions.
5. Traditional crop selection may depend mainly on experience.
6. Lack of proper agricultural information can affect crop selection.
7. Unsuitable crop selection can lead to low production.
8. Farmers may face financial losses due to poor crop decisions

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Who are the Stakeholders?

Internal Users:

1. System administrator
2. Agricultural experts
3. Data/system managers

External Users

1. Farmers
2. Agricultural consultants

Expected System Users

1. Farmers
2. Agricultural experts
3. System administrator

Objectives of the Project

1. To help farmers select suitable crops.
2. To analyse soil and environmental conditions.
3. To provide crop recommendations based on available data.
4. To reduce the chances of unsuitable crop selection.
5. To help farmers make better cultivation decisions.
6. To provide useful agricultural information.
7. To improve the efficiency of crop selection.
8. To support farmers in increasing the possibility of better crop production.

Expected Outputs

1. Farmer registration and profile management.
2. Collection of soil and environmental information.

3. Crop recommendation based on provided data.
4. Display of suitable crops.
5. Crop-related information and guidance.
6. Storage of farmer and recommendation records.
7. Agricultural reports and information.
8. Easy access to crop recommendations.

Hands-on Activity

Identify:

1. STRATEGIC INFORMATION

- Long-term crop production trends
- Regional crop trends
- Future agricultural planning

Example:
Agricultural experts can analyse long-term crop trends to support future agricultural planning.

2. TACTICAL INFORMATION

- Seasonal crop demand
- Crop performance during a season
- Regional crop recommendations
- Agricultural resource planning

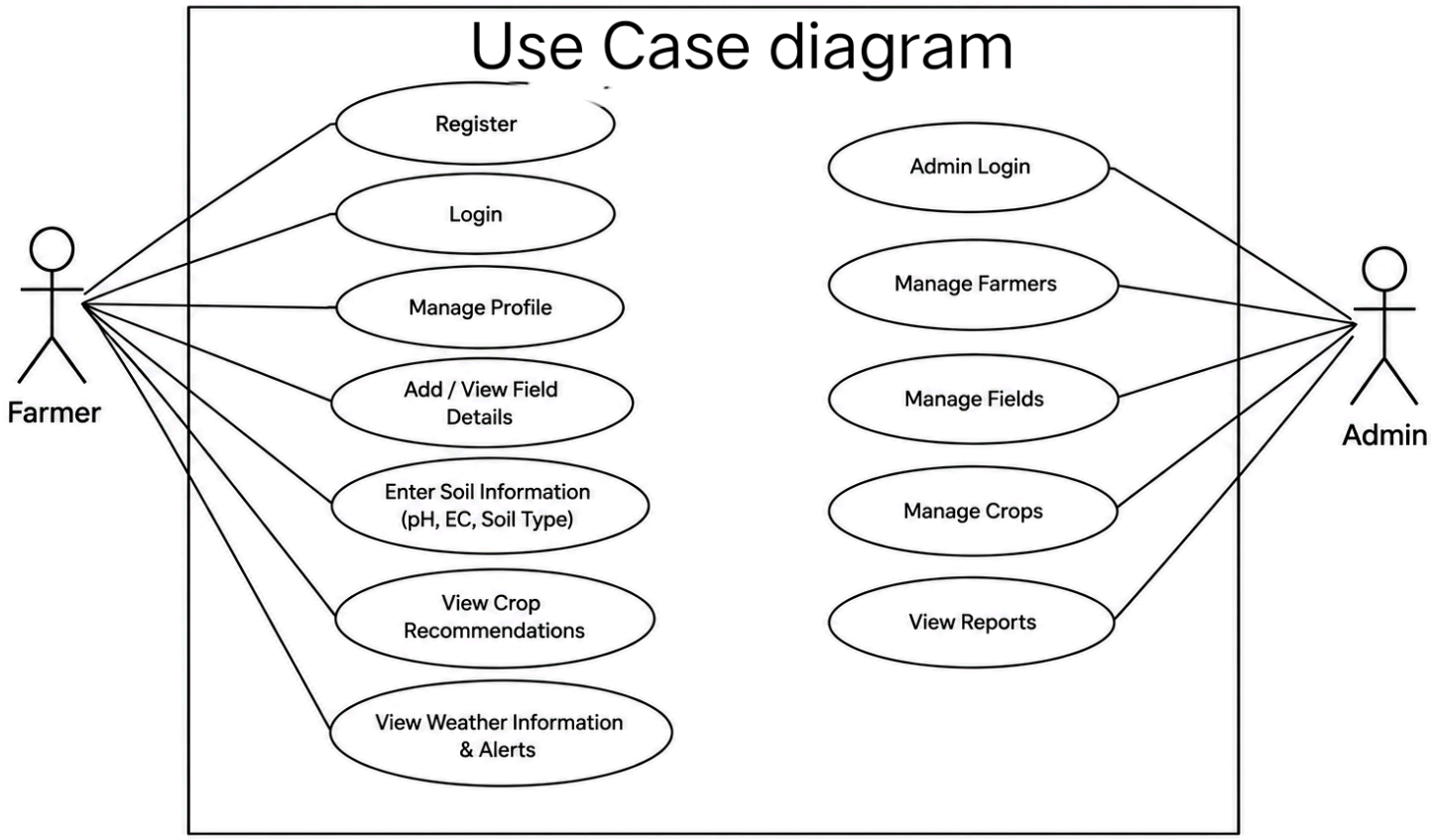
Example:
Agricultural experts can analyse seasonal information to determine which crops are suitable for a particular cultivation period.

3.OPERATIONAL INFORMATION

- Farmer details
- Current soil information

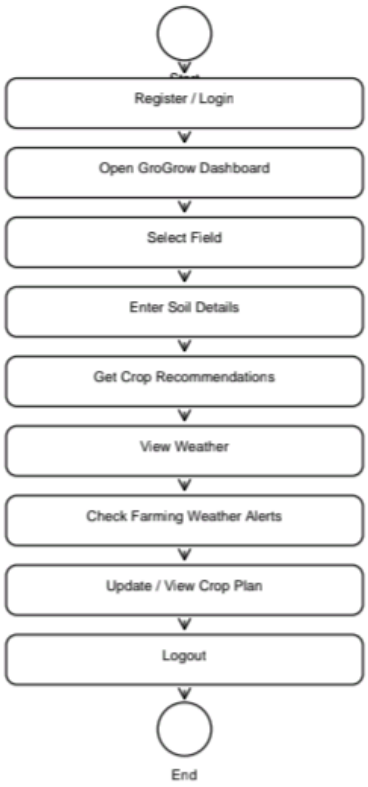
- Current environmental conditions
- Crop recommendation
- Daily agricultural information

Example:
A farmer can enter soil and environmental information into the system and receive a suitable crop recommendation.

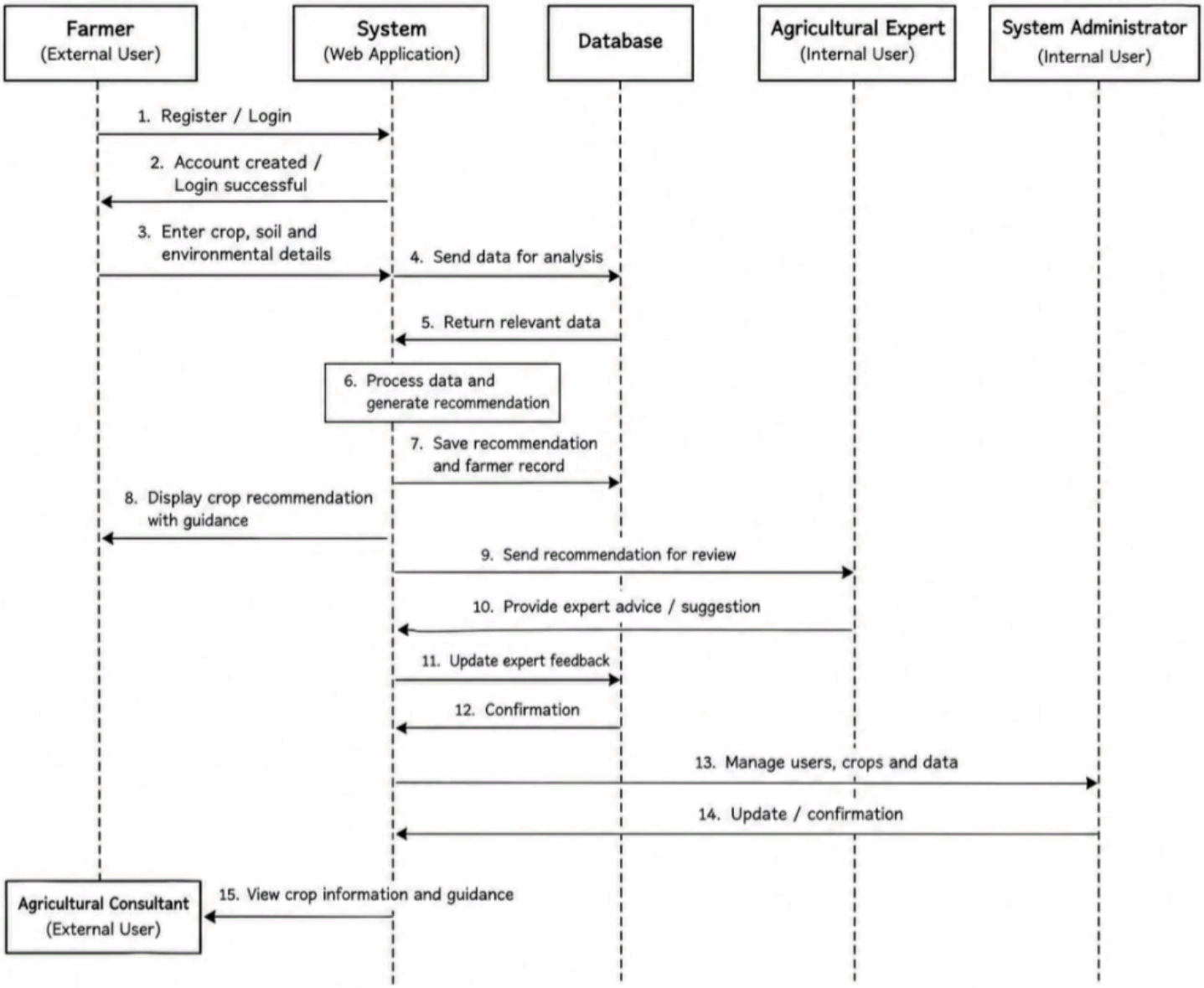

Use Case diagram
Register
Login
Manage Profile
Add / View Field Details
Enter Soil Information (pH, EC, Soil Type)
View Crop Recommendations
View Weather Information & Alerts
Admin Login
Manage Farmers
Manage Fields
Manage Crops
View Reports
Farmer
Admin

2. Activity Diagram

Farmer workflow in GroGrow

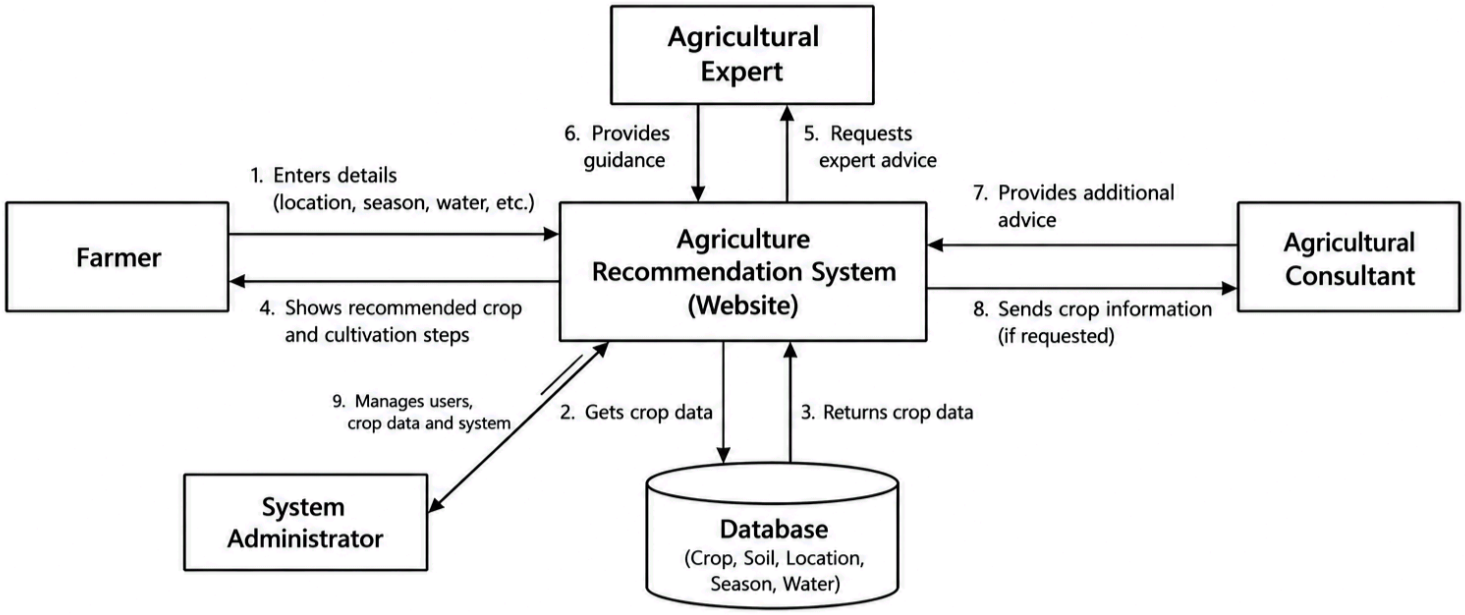


Sequential Diagram – Agriculture Crop Recommendation System (Crop Care)



Collaboration Diagram

(Agriculture Crop Recommendation System)



Class Diagram
(Agriculture Crop Recommendation System)

